

Tensor networks for neuro-symbolic AI

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- ▶ Tensor networks for neuro-symbolic AI

- ▶ Maximum entropy distributions

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Toy accounting rule system

Assume you want to book one of two accounts A1 and A2 depending on a feature F of an invoice, based on two rules:

- ▶ Exactly one account must be booked.
 - ▶ If feature F is present on the invoice, the account A1 is typically booked.
-
- ▶ Accounting rules are formalised in propositional logics, triggered by receipts properties.
 - ▶ These rules can either be hard restrictions, or prone to exceptions.

We use tensor networks to represent hard and soft accounting rules in a unified framework.

Tensors: Natural description of factored representations

Factored representations are collection of variables

$X_{[d]} = (X_0, \dots, X_{d-1})$, each with a finite set of states

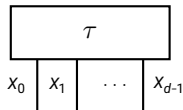
$[m_k] = \{0, \dots, m_k - 1\}$ [RN21]. The set of states is then the cartesian product

$$\bigtimes_{k \in [d]} [m_k] = \{0, \dots, m_0\} \times \dots \times \{0, \dots, m_{d-1}\}.$$

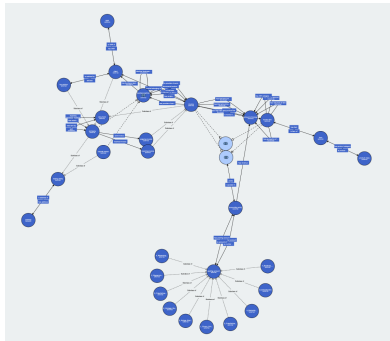
Knowledge about the system (possibility or uncertainty of a state) is a **tensor**

$$\tau [X_{[d]}] : \bigtimes_{k \in [d]} [m_k] \rightarrow \mathbb{R}.$$

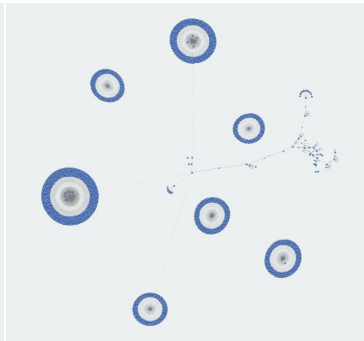
We represent tensors by diagrams with legs decorated by the variables:



The variables and main relations of the DATEV accounting ontology¹:



Coarse Visualization



Detailed Visualization

The curse of dimensionality prevents naive tensor representations.

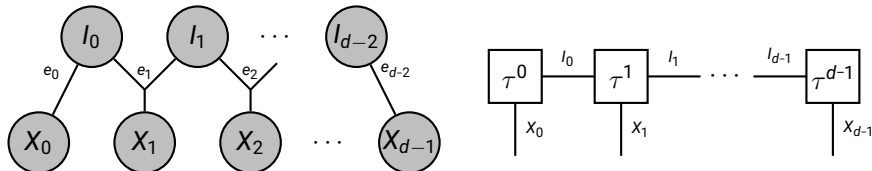
¹<https://github.com/EnexaProject/enexa-use-case-1-business-software-solutions>

Tensor Network Decompositions

To overcome the curse of dimensionality, tensors $\tau^{\mathcal{G}} [X_{\mathcal{V}}]$ are decomposed into core tensors $\tau^e [X_e]$. networks on hypergraphs $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

$$\tau^{\mathcal{G}} [X_{\mathcal{V}} = \mathbf{x}_{\mathcal{V}}] = \prod_{e \in \mathcal{E}} \tau^e [X_e = \mathbf{x}_e]$$

Example: TT format



Beyond black-boxes: Hypergraphs encode symbolic structure.

- Probabilistic perspective: **Conditional independencies**.
- Neural and logical perspective: **Function decompositions**.

Probabilistic Graphical models are Tensor Networks

Graphical models: Distributions are factorized into tensors $\tau^e [X_e]$, coordinate retrieval by

$$\mathbb{P} [X_{\mathcal{V}} = x_{\mathcal{V}}] = \prod_{e \in \mathcal{E}} \tau^e [X_e = x_e] .$$

This is the tensor decomposition

$$\mathbb{P} [X_{\mathcal{V}}] = \langle \{ \tau^e [X_e] : e \in \mathcal{E} \} \rangle_{[X_{\mathcal{V}}]} .$$

Interpretation of hypergraphs

Separation in hypergraphs implies conditional independencies in the distribution.

Example: Recommending a student

Consider a distribution of a student's intelligence X_I , SAT score X_S , test difficulty X_D , grade X_G and recommendation X_L .

The independencies are captured by:

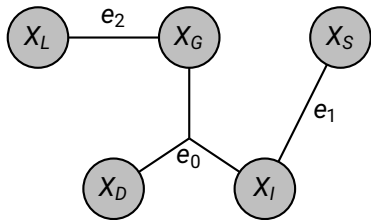
Let us assume the independencies:

- "The SAT score depends only on the student's intelligence":

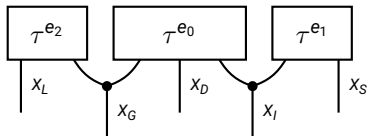
$$(X_S \perp X_{\{D,G,L\}}) \mid X_I$$

- "The recommendation letter depends only on the grade":

$$(X_L \perp X_{\{D,I,S\}}) \mid X_G$$



$\mathbb{P}[X_L, X_G, X_D, X_I, X_S]$ is decomposed by



Neural function decompositions

Example: Summation of integers in fixed-point representation

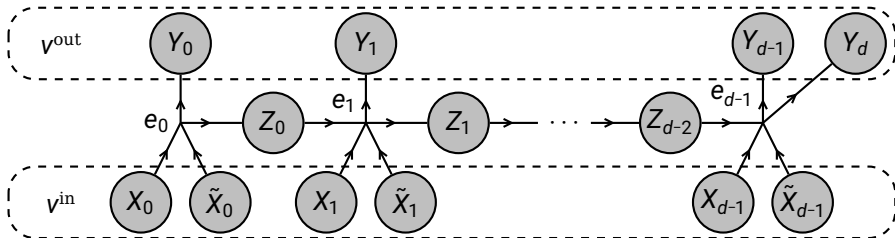
Any number $n \in [m^d]$ has a unique m -adic representation

$$\sum_{k \in [d]} x_k \cdot m^k$$

Given two numbers $x_{[d]}$ and $\tilde{x}_{[d]}$ we want the tuple $y_{[d+1]}$ such that

$$\sum_{k \in [d+1]} y_k \cdot m^k = \sum_{k \in [d]} (x_k + \tilde{x}_k) \cdot m^k.$$

The function decomposes into half-adders with carry bit variables $Z_{[d-1]}$:



Aim: Orient tensor network decompositions on circuit graphs.

We define the **basis encoding** of a function

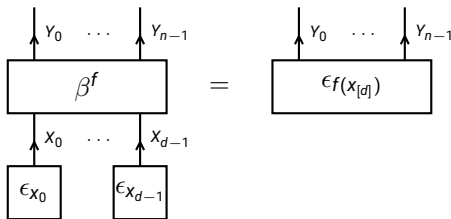
$$f : \bigtimes_{k \in [d]} [m_k] \rightarrow \bigtimes_{\ell \in [p]} [m_\ell]$$

as the tensor

$$\beta^f [Y_{[p]}, X_{[d]}] = \sum_{X_{[d]} \in \bigtimes_{k \in [d]} [m_k]} \epsilon_{f(X_{[d]})} [Y_{[p]}] \otimes \epsilon_{X_{[d]}} [X_{[d]}] .$$

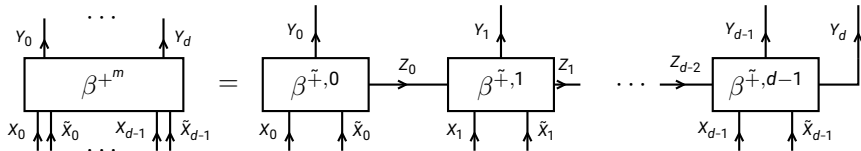
Basis calculus: Contraction with encoded input yields the encoded output.

$$\left\langle \beta^f [Y_{[p]}, X_{[d]}] , \epsilon_{X_{[d]}} [X_{[d]}] \right\rangle_{[Y_{[p]}]} = \epsilon_{f(X_{[d]})} [Y_{[p]}]$$

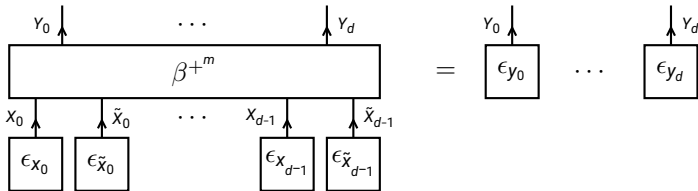


Example: Integer summation in fix-point representation

Orient on the circuit of half-adders we find a tensor network decomposition



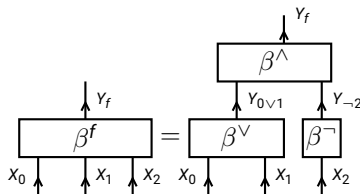
Basis Calculus: Contraction with one-hot encoded input integers yields the one-hot encoded sum.



Logical syntax translates in tensor network decompositions:

- Basis encodings of the logical connectives $\wedge, \vee, \neg$

Example: The basis encoding of $f[X_{[3]}] = (X_0 \vee X_1) \wedge \neg X_2$ is decomposed as:

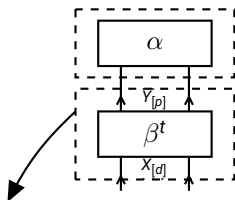


Outlook: First-order logics

Objects enumerated in additional tensor legs.

Computation-Activation Networks (CompActNets)

A novel tensor network architecture captures logical and probabilistic models (see [GSFE26]):



Activation of a statistic by an activation core, for **knowledge bases**

$$\alpha [Y_{[p]}] = \bigotimes_{\ell \in [p]} \begin{pmatrix} 0 \\ 1 \end{pmatrix} [Y_{\ell}]$$

and for **exponential families**

$$\alpha [Y_{[p]}] = \bigotimes_{\ell \in [p]} \begin{pmatrix} 1 \\ \exp [\theta [L = \ell]] \end{pmatrix} [Y_{\ell}]$$

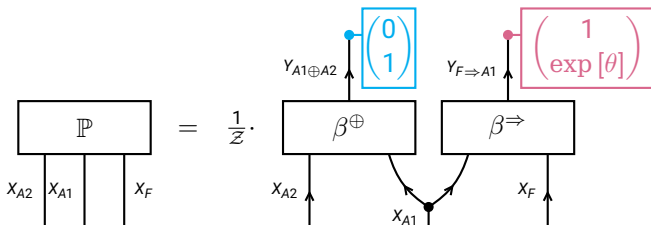
Computation of a statistic

$$t : \bigtimes_{k \in [d]} [m_k] \rightarrow \bigtimes_{\ell \in [p]} [p_{\ell}]$$

by basis encoding

Toy accounting system as a CompActNet

We represent the toy accounting system as a Computation-Activation Network:



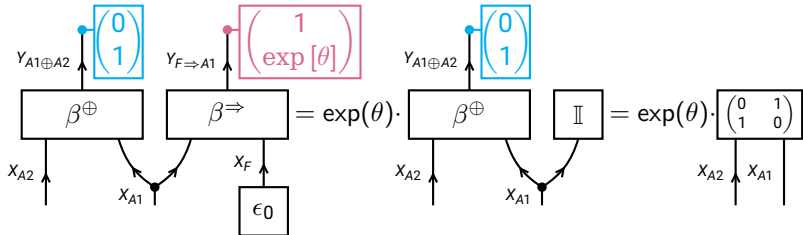
- ▶ The hard activation vector $\begin{pmatrix} 0 \\ 1 \end{pmatrix} [y_{A1 \oplus A2}]$ ensures that the first rule is always true.
- ▶ The soft activation vector $\begin{pmatrix} 1 \\ \exp[\theta] \end{pmatrix} [y_{F \Rightarrow A1}]$ introduces the trainable parameter $\theta \in \mathbb{R}$ controlling the satisfaction probability of the second rule.

Main inference principle

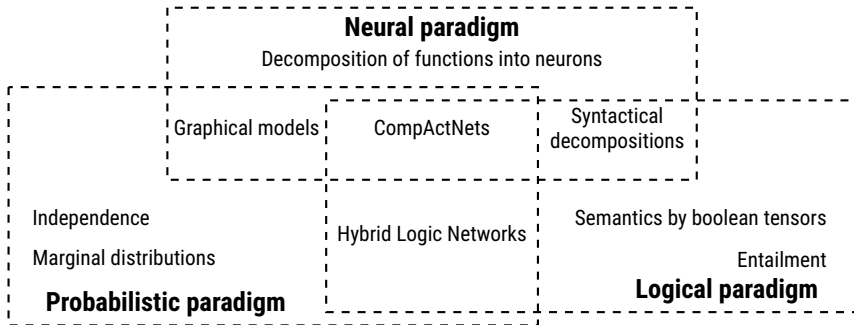
Model inference amounts to tensor network contractions.

- Probabilistic inference: Contractions are marginal probabilities.
- Logical inference: Contractions are model counts, which decide entailment.

Example: Distribution of X_{A1} and X_{A2} conditioned on $X_F = 0$ is parallel to



Tensor networks provide a unifying language to capture the different paradigms of AI [GSFE26]:



- ▶ Tensor networks for neuro-symbolic AI

- ▶ Maximum entropy distributions

Toy accounting rule system

Assume you want to book one of two accounts A1 and A2 depending on a feature F of an invoice, based on two rules:

- ▶ Exactly one account must be booked.
- ▶ If feature F is present on the invoice, the account A1 is typically booked.

Lets assume the following satisfaction rates of the toy accounting system on an invoice dataset:

$$\mu[L = 0] = 1$$

$$\mu[L = 1] = 0.75$$

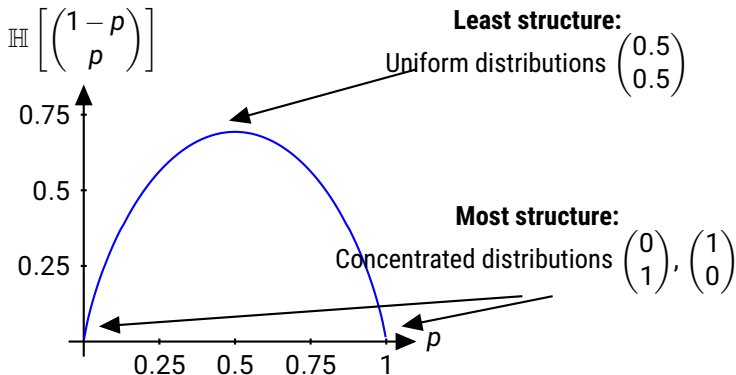
What is the "simplest" distribution reproducing these satisfaction rates?

Entropy: Quantifying the amount of structure

The entropy of a distribution $\mathbb{P} [X_{[d]}]$ is a measure of its structure:

$$\mathbb{H} [\mathbb{P} [X_{[d]}]] = - \sum_{x_{[d]} \in \times_{k \in [d]} [m_k]} \mathbb{P} [X_{[d]} = x_{[d]}] \cdot \ln [\mathbb{P} [X_{[d]} = x_{[d]}]]$$

Example: Entropy of Bernoulli distributions



Shannon Entropy: Natural measure of structure (negative information content) of a distribution.

Maximum entropy principle

Among those distributions reproducing certain characteristics of observed data, choose the one with the least structure (=maximum entropy).

To simplify notation we introduce selection encodings of functions t as tensors with coordinates

$$\sigma^t [X_{[d]} = x_{[d]}, L = \ell] = t_\ell(x_{[d]}).$$

Formally the maximum entropy problem is

$$\operatorname{argmax}_{\mathbb{P}[X_{[d]}]} \mathbb{H} [\mathbb{P} [X_{[d]}]] \quad \text{s.t.} \quad \langle \mathbb{P} [X_{[d]}], \sigma^t [X_{[d]}, L] \rangle_{[L]} = \mu [L].$$

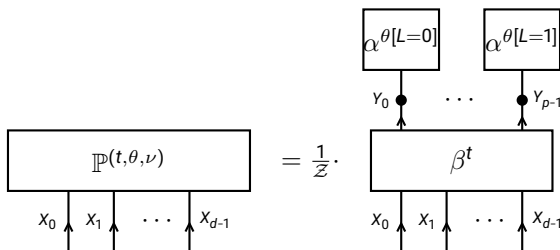
Exponential families are maximum entropy distributions

The exponential family of a statistic t is the set of distributions

$$\mathbb{P}^{(t, \theta)} [X_{[d]}] = \frac{1}{\mathcal{Z}(\theta)} \exp \left[\langle \theta [L], \sigma^t [X_{[d]}, L] \rangle_{[L]} \right]$$

where $\theta [L] \in \mathbb{R}^p$.

We find activation cores $\alpha^{\theta[L=\ell]} [Y_\ell]$ to represent the members as CompActNets:

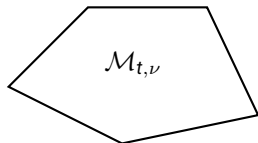


Classical result: Any member of the exponential family is a solution of the maximum entropy problem to its mean parameter $\mu [L]$ [KF09].

Realizable mean parameters form a convex polytope, the **mean polytope** of the statistic t :

$$\mathcal{M}_t = \text{conv} \left(\sigma^t [X_{[d]} = x_{[d]}, L] : x_{[d]} \in \prod_{k \in [d]} [m_k] \right)$$

Exponential families can only realize the relative interior of the mean polytope.



What are the solutions to the mean parameters on the boundary of the mean polytope?

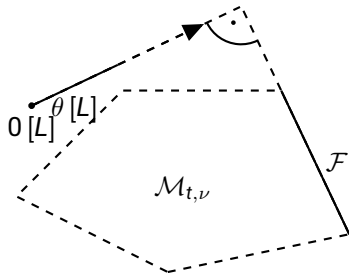
Mean parameters on the faces

Faces of the mean polytope are sets

$$\mathcal{F} = \operatorname{argmax}_{v \in \mathcal{M}_t} \langle \theta[L], v[L] \rangle_{[\emptyset]} \quad \mathbb{R}^p$$

We represent faces by activation cores [GE26]

$$\kappa^{\mathcal{F}} [Y_{[p]}] = \sum_{v \in \mathcal{F} \cap \times_{\ell \in [p]} [n_{\ell}]} \epsilon_v [Y_{[p]}]$$



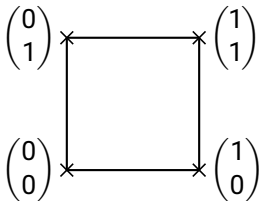
Theorem (Generic maximum entropy representation [GE26])

For any mean parameter on the relative interior of a face $\mathcal{F}$, we find a parameter $\theta[L]$ such that the corresponding maximum entropy is

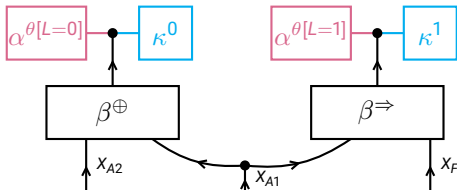
$$\frac{\langle \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}], \alpha^{\theta} [Y_{[p]}] \rangle_{[X_{[d]}]}}{\langle \beta^t [Y_{[p]}, X_{[d]}], \kappa^{\mathcal{F}} [Y_{[p]}], \alpha^{\theta} [Y_{[p]}] \rangle_{[\emptyset]}}$$

Toy accounting example

The mean polytope is

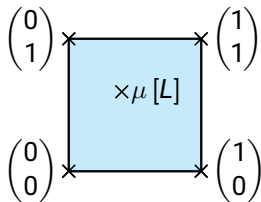


Any solution of the maximum entropy problem is a CompActNet with elementary activation.

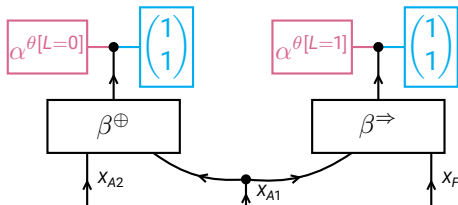


Toy accounting example

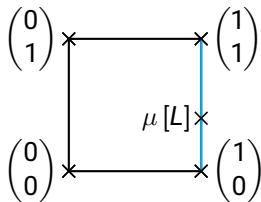
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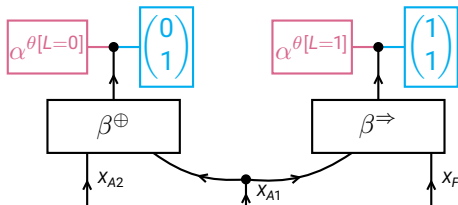
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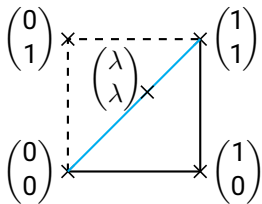
Any solution of the maximum entropy problem is a CompActNet with elementary activation.



Consider the statistic

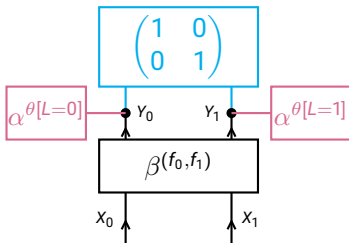
$$f_0[X_0, X_1] = (X_0 \wedge X_1) \quad , \quad f_1[X_0, X_1] = (X_0 \Rightarrow X_1)$$

The mean polytope of this statistic is



The blue face cannot be represented by an elementary tensor.

Representation of the maximum entropy distribution for a mean parameter in the blue face:



Thank you very much for your attention!

- [GE26] Alex Goessmann and Martin Eigel.
Tensor network representations of discrete maximum entropy distributions via mean polytopes, September 2026.
arXiv:2609.07184 [math.ST].

- [GSFE26] Alex Goessmann, Janina Schütte, Maximilian Fröhlich, and Martin Eigel.
A tensor network formalism for neuro-symbolic AI, January 2026.
arXiv:2601.15442 [cs].

- [KF09] Daphne Koller and Nir Friedman.
Probabilistic Graphical Models: Principles and Techniques.
The MIT Press, Cambridge, Mass., 1. edition edition, July 2009.

- [RN21] Stuart Russell and Peter Norvig.
Artificial Intelligence: A Modern Approach, Global Edition: A Modern Approach, Global Edition.
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